

Data Structures : Graph

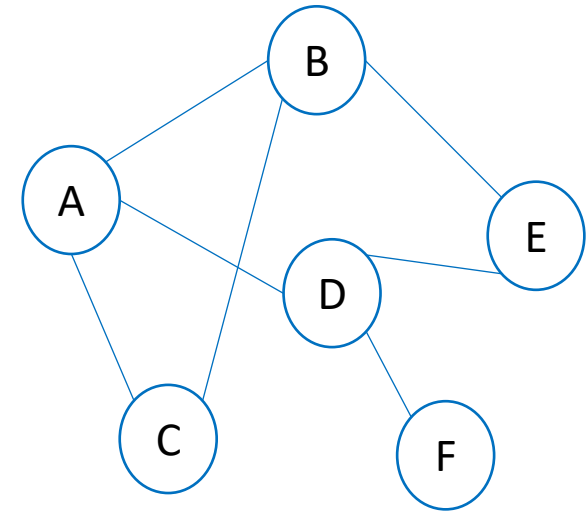
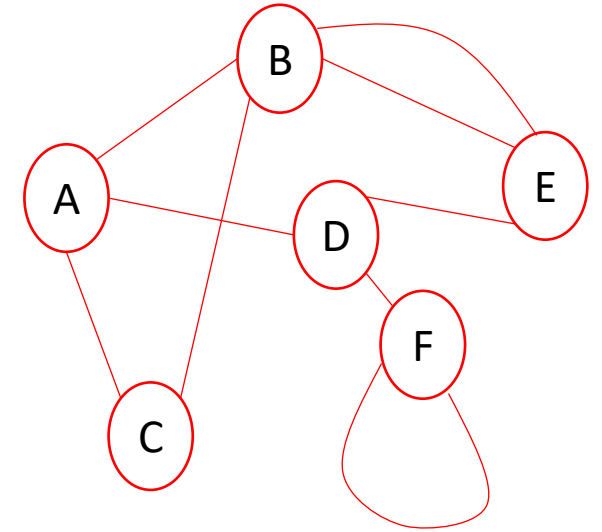
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Graph

- Graph is a non-linear data structure.
- Graph is defined as set of vertices and edges. Vertices (also called as nodes) hold data, while edges connect vertices and represent relations between them.
 - $G = \{ V, E \}$
- When there is an edge from vertex P to vertex Q, P is said to be adjacent to Q.
- Multi-graph
 - Contains multiple edges in adjacent vertices or loops (edge connecting a vertex to it-self).
- Simple graph
 - Doesn't contain multiple edges in adjacent vertices or loops.

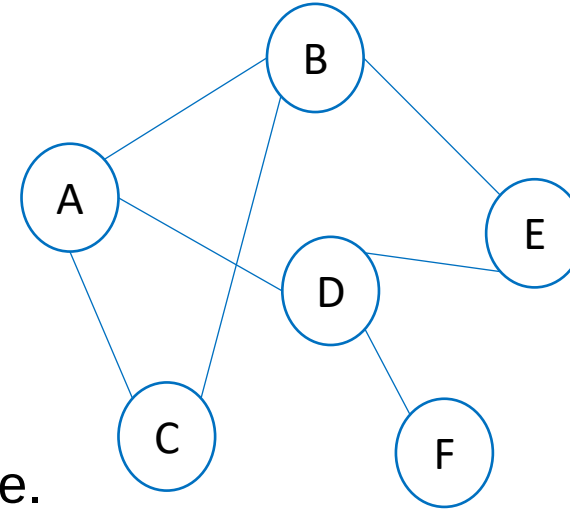


Graph

- Graph edges may or may not have directions.

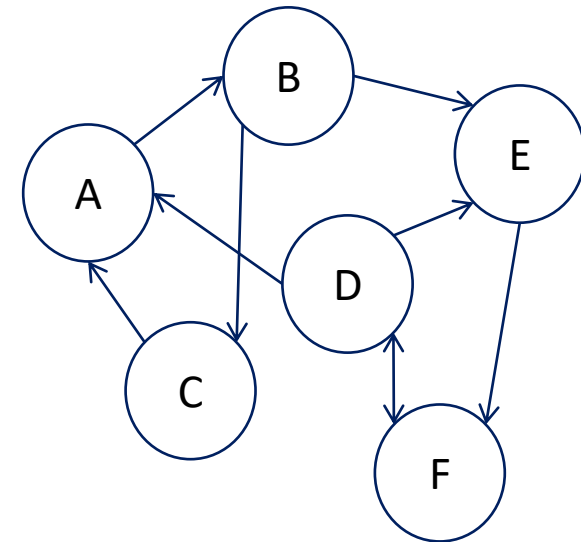
- Undirected Graph: $G = \{ V, E \}$

- $V = \{ A, B, C, D, E, F \}$
- $E = \{ (A,B), (A,C), (A,D), (B,C), (B,E), (D,E), (D,F) \}$
- If P is adjacent to Q, then Q is also adjacent to P.
- Degree of node: Number of nodes adjacent to the node.
- Degree of graph: Maximum degree of any node in graph.



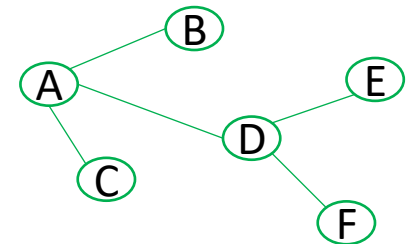
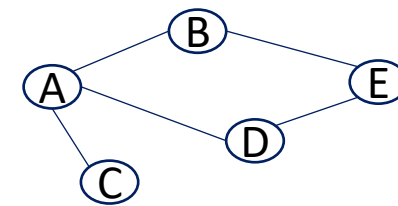
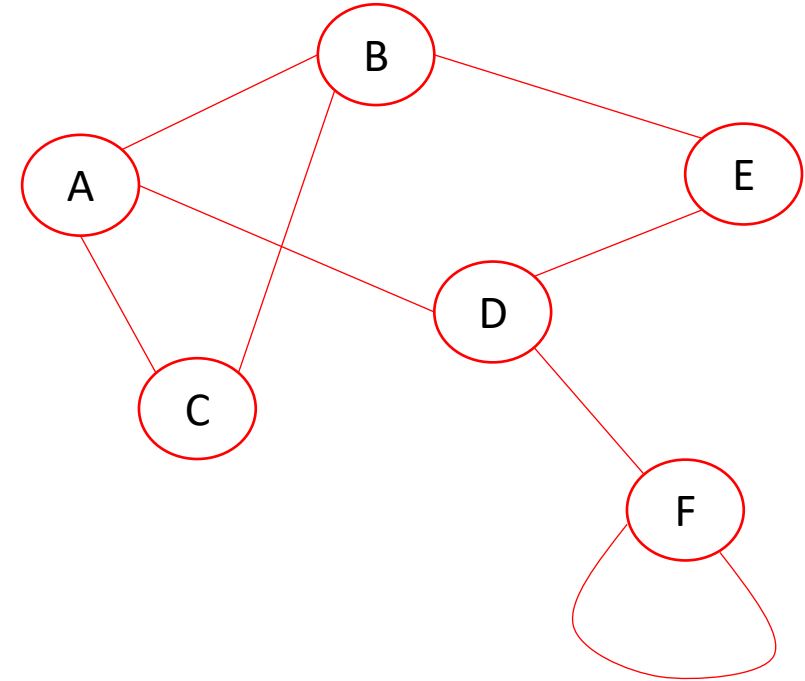
- Directed Graph: $G = \{ V, E \}$

- $V = \{ A, B, C, D, E, F \}$
- $E = \{ \langle A,B \rangle, \langle B,C \rangle, \langle B,E \rangle, \langle C,A \rangle, \langle D,A \rangle, \langle D,E \rangle, \langle D,F \rangle, \langle E,F \rangle, \langle F,D \rangle \}$
- If P is adjacent to Q, then Q is may or may not be adjacent to P.
- Out-degree: Number of edges originated from the node
- In-degree: Number of edges terminated on the node



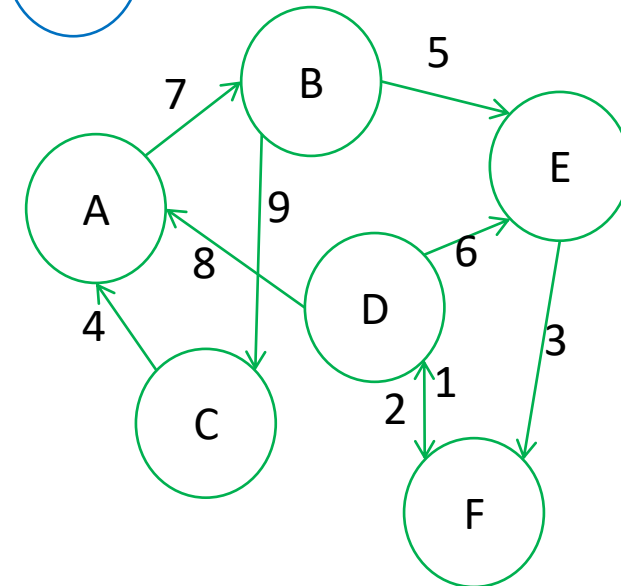
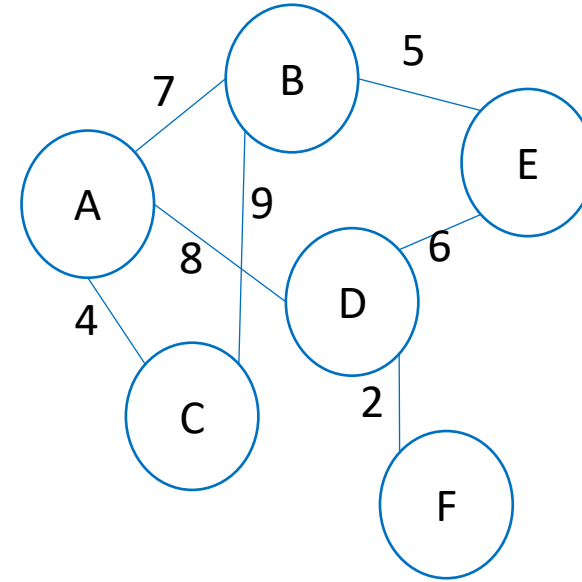
Graph

- Path: Set of edges between two vertices. There can be multiple paths between two vertices.
 - A – D – E
 - A – B – E
 - A – C – B – E
- Cycle: Path whose start and end vertex is same.
 - A – B – C – A
 - A – B – E – D – A
- Loop: Edge connecting vertex to itself. It is smallest cycle.
 - F – F
- Sub-Graph: A graph having few vertices and few edges in the given graph, is said to be sub-graph of given graph.



Graph

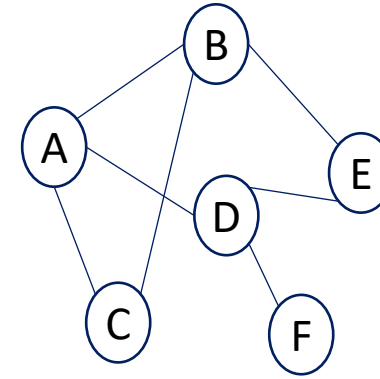
- **Weighted graph**
 - Graph edges have weight associated with them.
 - Weight represent some value e.g. distance, resistance.
- **Directed Weighted graph (Network)**
 - Graph edges have directions as well as weights.
- **Applications of graph**
 - Electronic circuits
 - Social media
 - Communication network
 - Road network
 - Flight/Train/Bus services
 - Bio-logical & Chemical experiments
 - Deep learning (Neural network, Tensor flow)
 - Graph databases (Neo4j)



Graph

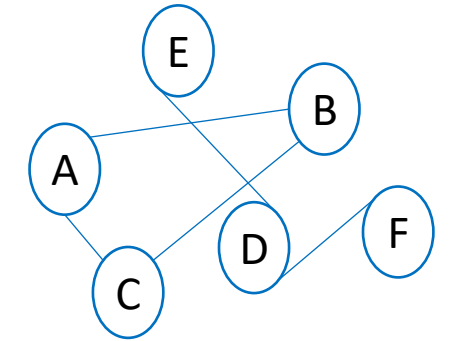
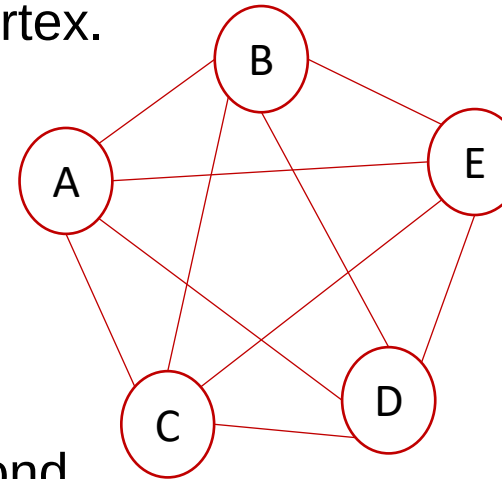
- **Connected graph**

- From each vertex some path exists for every other vertex.
- Can traverse the entire graph starting from any vertex.



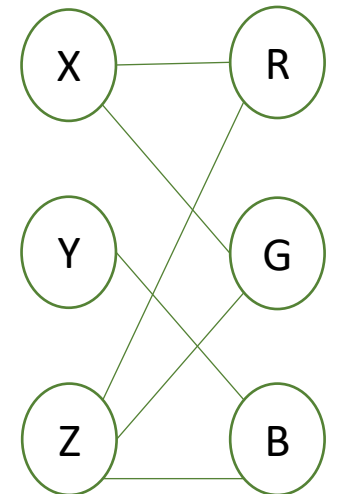
- **Complete graph**

- Each vertex of a graph is adjacent to every other vertex.
- Un-directed graph: Number of edges = $n(n-1) / 2$
- Directed graph: Number of edges = $n(n-1)$



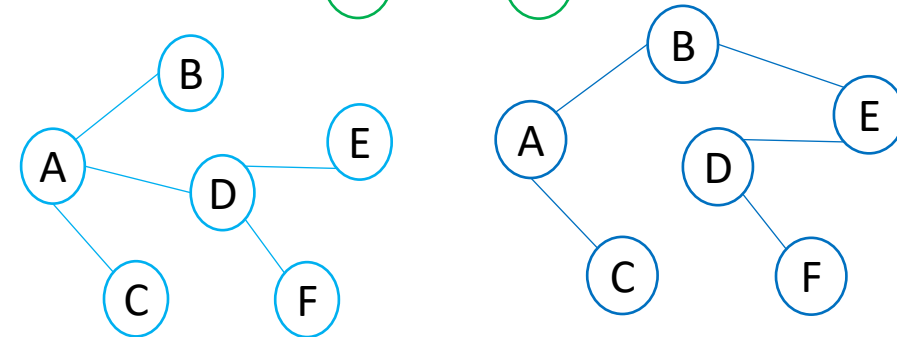
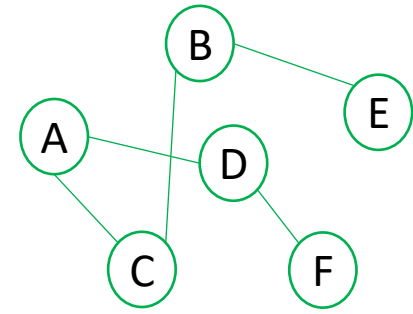
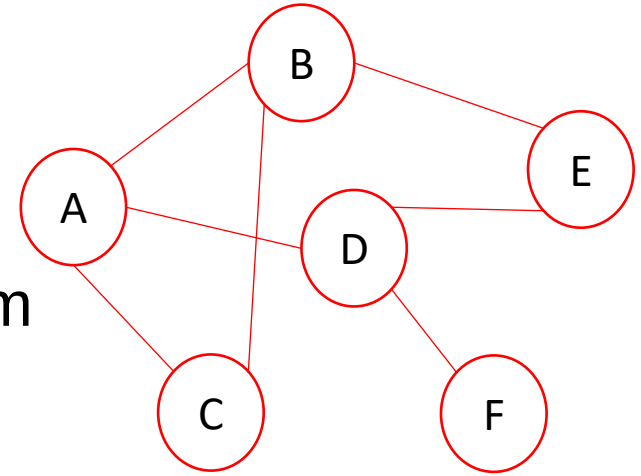
- **Bi-partite graph**

- Vertices can be divided in two disjoint sets.
- Vertices in first set are connected to vertices in second set.
- Vertices in a set are not directly connected to each other.



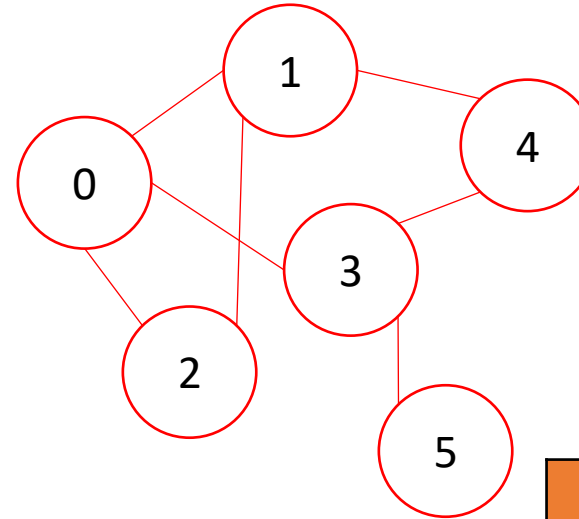
Spanning Tree

- Tree is a graph without cycles.
- Spanning tree is connected sub-graph of the given graph that contains all the vertices and sub-set of edges ($V-1$).
- Spanning tree can be created by removing few edges from the graph which are causing cycles to form.
- One graph can have multiple different spanning trees.
- In weighted graph, spanning tree can be made who has minimum weight (sum of weights of edges). Such spanning tree is called as Minimum Spanning Tree.
- Spanning tree can be made by various algorithms.
 - BFS Spanning tree
 - DFS Spanning tree
 - Prim's MST
 - Kruskal's MST



Graph Implementation – Adjacency Matrix

- If graph have V vertices, a $V \times V$ matrix can be formed to store edges of the graph.
- Each matrix element represent presence or absence of the edge between vertices.
- For non-weighted graph, 1 indicate edge and 0 indicate no edge.
- For un-directed graph, adjacency matrix is always symmetric across the diagonal.
- Space complexity of this implementation is $O(V^2)$.

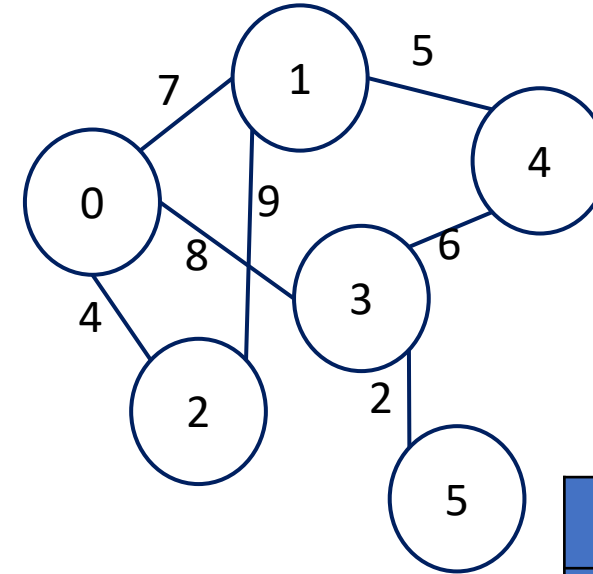


	0	1	2	3	4	5
0						
1						
2						
3						
4						
5						



Graph Implementation – Adjacency Matrix

- If graph have V vertices, a $V \times V$ matrix can be formed to store edges of the graph.
- Each matrix element represent presence or absence of the edge between vertices.
- For weighted graph, weight value indicate the edge and infinity sign ∞ represent no edge.
- For un-directed graph, adjacency matrix is always symmetric across the diagonal.
- Space complexity of this implementation is $O(V^2)$.

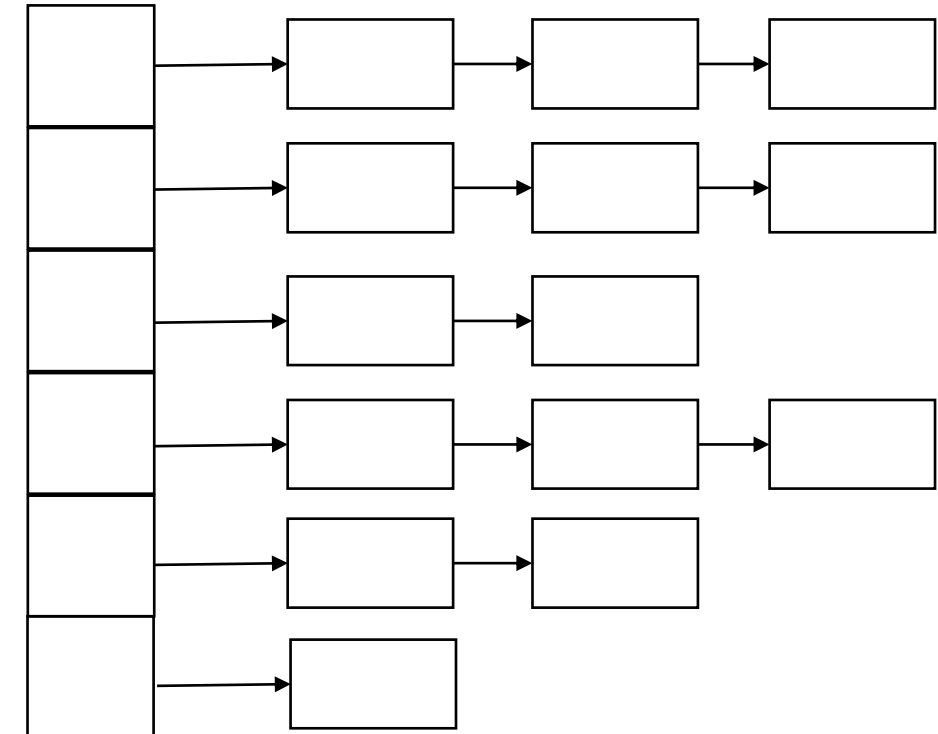
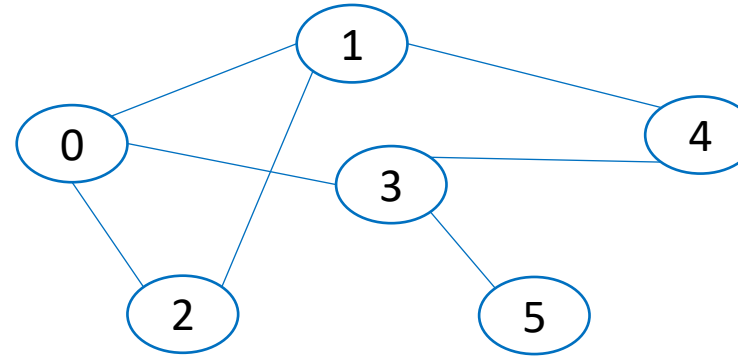


	0	1	2	3	4	5
0						
1						
2						
3						
4						
5						



Graph Implementation – Adjacency List

- Each vertex holds list of its adjacent vertices.
- For non-weighted graphs only, neighbour vertices are stored.
- For weighted graph, neighbour vertices and weights of connecting edges are stored.
- Space complexity of this implementation is $O(V+E)$.
- If graph is sparse graph (with fewer number of edges), this implementation is more efficient (as compared to adjacency matrix method).





Thank you!

